

AUTONOMOUS PARKING PROJECT: INTERFACE DESIGN & UNIT TESTING REPORT

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1. Introduction

This report outlines all the activities involved in designing and testing an autonomous parking assistance system. We begin by describing our understanding of the overall system, outlining any assumptions made in the requirements and any extra assumptions we came up with. We then outline the functional and non-functional requirements and use that to describe the initial system interface. Once we have the requirements and the system interface, we proceed to design each test case. Finally, we use the interface, the designed test cases to iteratively implement test cases as well as the needed functionality in each interface method.

1.1. System description

The system in question involves aiding an autonomous car to park in parking spots available in a perfectly straight parking stretch of 500m. The car moves in a perfectly straight line in this street, and the parking spots are located to the right-hand side of said street. The car is also equipped with two sensors to detect any obstacles to the right-hand side of the car.

The car's parking functionality is described as follows:

- a. Move forward by 1 meter.
- b. Query the sensors to check if there is any obstacle to the right.
- c. Track whether the space to the right is free or not.
- d. Repeat step a, b and c until a free stretch of 5m is found.
- e. If a free stretch of 5m is found, move to the end of the 5m stretch and perform a. reverse parking maneuver

1.2. Assumptions

The provided project document states the following base assumptions:

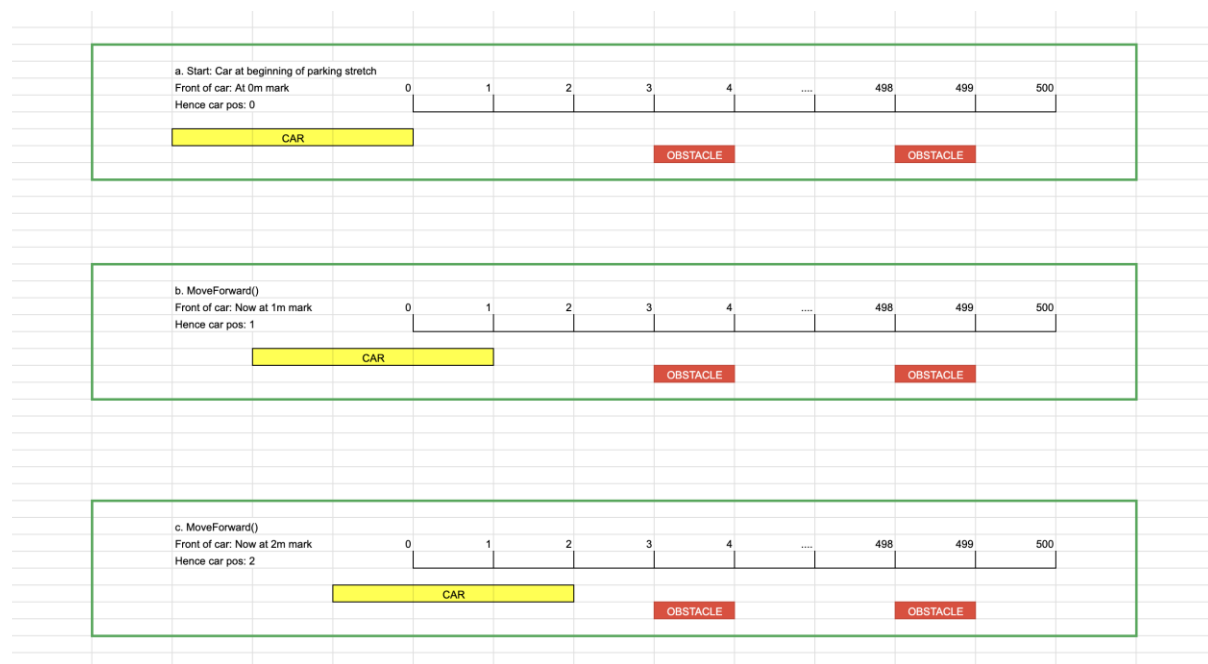
1. The parking stretch is perfectly straight and is 500 meters long.
2. The car moves forward (or backwards) in a perfectly straight manner in exactly 1-meter steps.
3. The car needs at least 5m to park.
4. The car's sensor report values between 0 and 200 cm.
5. The distance in centimetre which sensors reports an object-free space is 150 centimetres.
6. The noise sensor value is considered as the difference between the maximum and minimum value of each time sensor is queried is 80 cm.

We've added our own assumptions to bound the problem description and simplify the parts of the system we would need to test:

1. We assume the actual parking movements/routines to be out of scope; thus, we just need to state the state transitions from parked to unparked and vice versa.
2. We assume the car in question is the only moving object in the duration of our tests
3. The car is less than 5m long, this is to leave ample space for the car to realistically reverse park and unpark

4. We measure the car's position using the front of the car i.e. if the front is at the 1m mark of the stretch, then we say the position = 1
5. The obstacles in the parking spots, if any, are static
6. The sensors are located at the front of the car, such that they detect obstacles directly to the right of the front of the car
7. The car requires at least 150cm to the right so as to park without blocking the street
8. The obstacles in the parking spots are at least 1m long, and the front of each obstacle is situated at the end of (the first) 1m spot that the obstacle occupies. Thus, we can assume that an entire 1m spot is occupied if our sensors report any obstacles closer than 150cm to the right

An example of the 500m street, the car's movement as well as obstacles are visualized in the diagram below:



2. Functional Requirements Analysis

- Below are the functional requirements broken down from the general requirements of the project:

Req No	Name	Type	Function Relevance	Descriptions
R1.1	Street and position range	Functional	MoveForward(), MoveBackward(), WhereIs()	0 m <= Car position
R1.2		Functional	MoveForward(), MoveBackward(), WhereIs()	Car position <= 500 m
R1.3		Non-Functional	MoveForward(), MoveBackward()	Free parking distance valid detected by sensors >= 150 cm
R2.1	Two ultrasound sensors range	Functional	isEmpty()	0 cm <= Sensor data
R2.2		Functional	isEmpty()	Sensor data <= 200 cm
R3.1	Read Sensor Data	Functional	isEmpty()	Query both sensors
R3.2		Functional	isEmpty()	Query them at least 5 times.
R3.3		Functional	isEmpty()	Filter noise
R3.4		Functional	isEmpty()	Return the distance to the nearest object on the right-hand side (minimum of the two sensors)
R4.1	Faulty/noisy sensor	Functional	isEmpty()	Both sensors normal
R4.2		Functional	isEmpty()	Sensor 1 noisy
R4.3		Functional	isEmpty()	Sensor 2 noisy
R4.4		Functional	isEmpty()	Both noisy
R4.5		Non-Functional	isEmpty()	Filter value deviation <= 80 cm, sensor data is valid
R4.6		Non-Functional	isEmpty()	Filter value deviation > 80 cm, sensor data is invalid
R5.1	Parking-space detection	Functional	Park()	< 5 m (not sufficient)
R5.2		Functional	Park()	is 5 m (free)
R5.3		Functional	Park()	> 5 m (not sufficient)
R6.1	Move forward	Functional	MoveForward()	Move 1 m forward per step
R6.2		Functional	MoveForward()	Query the sensors using isEmpty()
R6.3		Functional	MoveForward()	Return Data Structure (Current Car Position + Current Parking Status)
R6.4		Functional	MoveForward()	Cannot move if current position is 500m
R7.1	Move backward	Functional	MoveBackward()	Move 1 m backward per step

R7.2		Functional	MoveBackward()	Query the sensors using isEmpty()
R7.3		Functional	MoveBackward()	Return Data Structure (Current Car Position + Current Parking Status)
R7.4		Functional	MoveBackward()	Cannot move if current position is 0m
R8.1		Functional	Park()	Detect free space -> Park
R8.2	Parking	Functional	Park()	No available space -> Move Forward -> Detect free space -> Park
R8.3		Functional	Park()	No available space -> Move Forward -> No available space till end of the stretch
R9.1	Unparking	Functional	UnPark()	Car is Parked -> Unparked
R9.2		Functional	UnPark()	Car is Unparked -> Unparked
R10.1	Where Is	Functional	WhereIs()	Return Current position
R10.2		Functional	WhereIs()	Return Parking status
R11	Actuators	Non-Functional		Not Modelled
R12	Sensors	Non-Functional		Not Modelled

3. Interface Design

- AutonomousParking is the main controller, implementing the AutonomousParkingInterface.
- It interacts with two sensors through IDataSensor interface
- DataSensor provides the sensor-data processing functions, implementing the IDataSensor interface.
- The system maintains the current car position, parking status, and detected free-space length – current system state.
- Other functions such as MoveForward(), MoveBackward(), IsEmpty(), Park(), UnPark(), and WhereIs() are used to process and return the current system state as mentioned before.

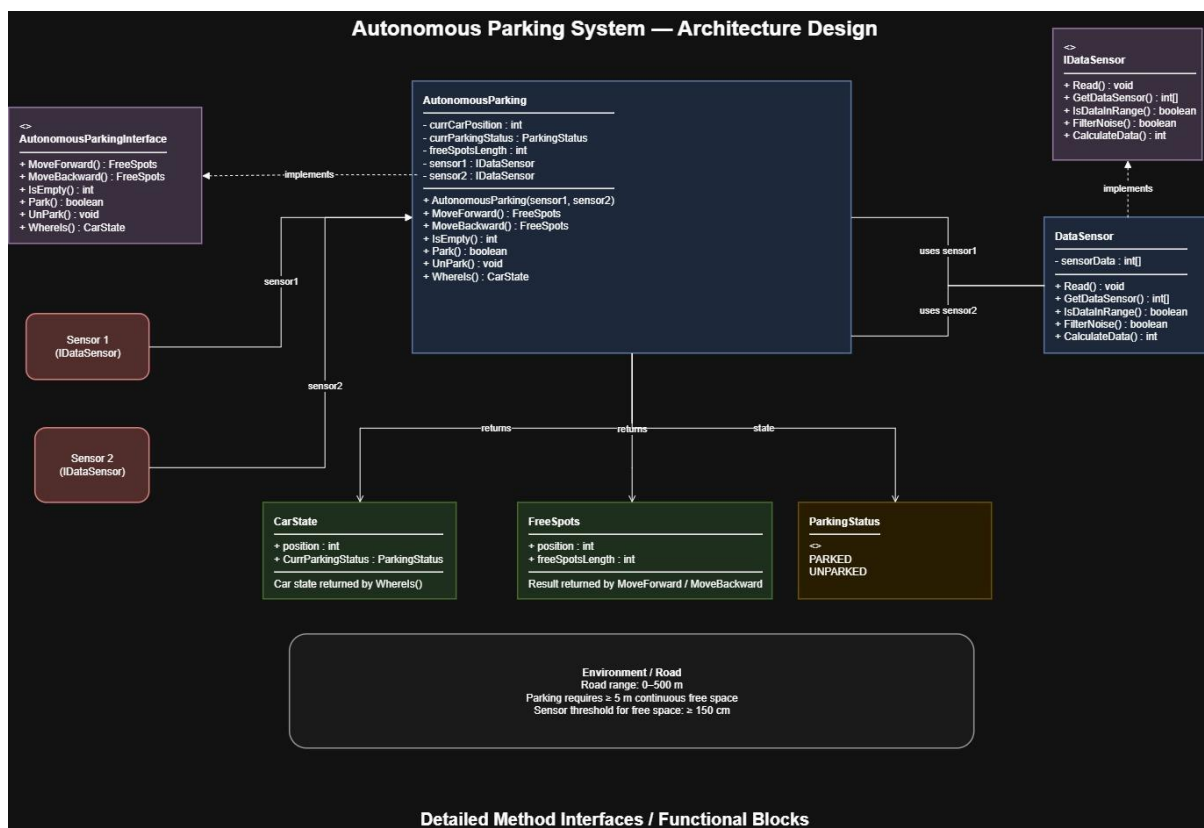


Figure 3.1: Interface design of the Autonomous Parking System

4. Test cases design

For each interface method, we state/link the intended functional requirements as well the required input and outputs to the method. We also list the invariants that should hold before and after the procedure is executed. Lastly, we describe the test cases for the method using a specific design method and state our reasoning for picking said methodology.

4.1. MoveForward

4.1.1. Requirements

Requirements R1.1, R1.2, R1.3, R6.1, R6.2, R6.3 and R6.4 (listed in [Section 2](#)) are covered this method.

4.1.2. Inputs

- Queries the car's position from internal state via the WhereIs() method.
- Queries for the distance to closest object on the right using the IsEmpty() method.

4.1.3. Outputs

- Updates the car's internal position state
- Returns the updated car's position and the distance/no. of free parking spaces detected thus far.

4.1.4. Pre-Conditions

The following invariants should hold before MoveForward() is executed:

- The car's position is greater than or equal to 0 but less than or equal to 499
- The car is in an unparked state

4.1.5. Post-Conditions

The invariants below should hold after MoveForward() is executed:

- The car's position should increment by 1 i.e. be greater than or equal to 1 but less than or equal to 500
- The car should still be in an unparked state

4.1.6. Test design technique

A decision table was selected because the logic of the function branches in a binary manner depending on the state of the car. Using this technique also make it easy to match the car's conditions against expected actions/outcomes after the MoveForward() method is executed:

Conditions / Actions	TC-MF-1	TC-MF-2	TC-MF-3	TC-MF-4
c1: $0 \leq \text{position} \leq 499$	True	True	True	False
c2: $\text{carParkingState} = \text{UNPARKED}$	True	True	False	-
c3: $\text{isEmpty}() \geq 150\text{cm}$	True	False	-	-
a1: wrong input/state	-	-	X	X
a2: $\text{position} += 1, \text{freeSpots} = 0$	-	X	-	-
a3: $\text{position} += 1, \text{freeSpots} += 1$	X	-	-	-

4.2. IsEmpty:

4.2.1. Requirements:

Requirements R2.1, R2.2, R3.1, R3.2, R3.3, R3.4, R4.1, R4.2, R4.3, R4.4, R4.5 and R4.6 (listed in [Section 2](#)) are covered in this method.

4.2.2. Inputs

- SENSOR_DATA1 = {data_1, data_2, data_3, data_4, data_5}
- SENSOR_DATA2 = {data_1, data_2, data_3, data_4, data_5}

4.2.3 Outputs

- Return a filtered distance value in centimetres

4.2.4. Pre-conditions

- Both sensors data are available
- Sensor data range must be ≥ 0 and ≤ 200
- The sensors can be queried at least 5 times.

4.2.5. Post-conditions

- If both sensors are invalid, return -1
- If one sensor is valid, return its filtered data
- If both are valid, return the minimum filtered data of them

4.2.6. Assumptions

- Sensor data will be given as an array of 5 elements for at least 5 times query
- A sensor is considered invalid if its readings have a deviation greater than 80 cm.
Deviation = Max reading - Min reading
- Method to calculate overall sensor data is an average of 5 times

4.2.7. Test design technique

4.2.7.1. Decision Tables

A decision table was selected because the behaviour of IsEmpty() depends on the combination of the validity states of the two sensors. Each sensor can be either valid or invalid, resulting in four possible combinations. The decision table ensures that all combinations are covered and that the corresponding output (min(avg(S1), avg(S2)), S1, S2, or -1) is explicitly defined.

Conditions / Actions	TC_IE_01	TC_IE_02	TC_IE_03	TC_IE_04
c1: S1 state	Valid	Valid	Invalid	Invalid
c2: S2 state	Valid	Invalid	Valid	Invalid
a1: IsEmpty() = Min(avg(S1), avg(S2))	X	-	-	-
a2: IsEmpty() = S1	-	X	-	-
a3: IsEmpty() = S2	-	-	X	-
a4: IsEmpty() = -1	-	-	-	X

4.2.7.2. Boundary Value Analysis

Boundary value analysis was selected because the sensor input has clearly defined valid limits of 0 and 200. Values immediately below, at, and above these boundaries are tested to verify that the implementation correctly distinguishes valid sensor readings from out-of-range readings.

Average Input (5 times)				
Conditions / Actions	TC_IE_05	TC_IE_06	TC_IE_07	TC_IE_08
c1: S1 average value	< 0	0	200	> 200
c2: S2 average value	< 0	0	200	> 200
a3: Sensor state	Invalid	Valid	Valid	Invalid
a1: IsEmpty() = -1	X	-	-	X
a2: IsEmpty() = Min(avg(S1), avg(S2))	-	X	X	-

Boundary value analysis was also applied to the noise/deviation threshold because the sensor is considered invalid when the deviation exceeds 80. Testing values below, equal to, and above 80 verifies the behaviour at this critical decision boundary.

Noise (Deviation = 80)			
Conditions / Actions	TC_IE_09	TC_IE_10	TC_IE_11
c1: S1 average value	< 80	80	> 80
c2: S2 average value	< 80	80	> 80
a3: Sensor state	Invalid	Valid	Invalid
a1: IsEmpty() = -1	X	-	X
a2: IsEmpty() = Min(avg(S1), avg(S2))	-	X	-

4.3. MoveBackward

4.3.1. Requirements

Requirements R1.1, R1.2, R1.3, R7.1, R7.2, R7.3 and R7.4 (listed in [Section 2](#)) are satisfied by this method.

4.3.2. Inputs

- Queries the car's position from internal state via the WhereIs() method
- Queries for the distance to closest object on the right using the IsEmpty() method

4.3.3. Outputs

- Updates the car's internal position state
- Returns the updated car's position and the distance/no. of free parking spaces detected thus far

4.3.4. Pre-Conditions

The following invariants should hold before MoveBackward() is executed:

- The car's position is greater than or equal to 1 but less than or equal to 500
- The car is in an unparked state

4.3.5. Post-Conditions

The following invariants should hold after MoveBackward() is executed:

- The car's position decrement by 1 i.e. be greater than or equal to 0 but less than or equal to 499
- The car should still be in an unparked state

4.3.6. Test design technique

Like MoveForward(), we chose a decision table to model the test cases for the same reasons.

Conditions / Actions	TC-MB-1	TC-MB-2	TC-MB-3	TC-MB-4
c1: $1 \leq \text{position} \leq 500$	True	True	True	False
c2: $\text{carParkingState} = \text{UNPARKED}$	True	True	False	-
c3: $\text{isEmpty}() \geq 150\text{cm}$	True	False	-	-
a1: wrong input/state	-	-	X	X
a2: $\text{position} -= 1, \text{freeSpots} = 0$	-	X	-	-
a3: $\text{position} -= 1, \text{freeSpots} += 1$	X	-	-	-

4.4. Park

4.4.1. Requirements:

Requirements R5.1, R5.2, R5.3, R8.1, R8.2, R8.3 (listed in [Section 2](#)) are covered in this method.

4.4.2. Inputs

- Current CarPosition
- The number of free spots

4.4.3 Outputs

- Return a boolean value
 - True: Car is parked successfully
 - False: Car is not parked successfully

4.4.4. Pre-conditions

- Car's parking status is currently UNPARKED
- Car's position is in the range between 0 and 500 meters of the stretch

4.4.5. Post-conditions

- If a suitable free parking stretch of at least 5 meters is detected:
 - The car moves to the end of the stretch and drops at this parking stretch on the way.
 - The pre-programmed reverse parallel parking maneuver is performed.
 - The car's parking status becomes PARKED.

- If no suitable parking stretch is detected before the end of the street:
 - The car's position is 500 meters.
 - The car's parking status remains UNPARKED.

4.4.6. Assumptions

- A valid parking space is a continuous free stretch of at least 5 meters.
- The car detects parking availability while moving forward using isEmpty().
- The car is considered to be at a ready parking space when a valid 5-meter free stretch has been detected.
- The parking maneuver is pre-programmed and does not require sensor feedback during the maneuver.
- The actuator signals used to perform the parking maneuver are not modeled in this phase.
- If no suitable parking space is found before the end of the street, the car remains unparked.

4.4.7. Test design technique

4.4.7.1. Decision Tables

A decision table is chosen for this method because Park behaves depending on the several logical combinations. The method determines whether the car is already at the parking stretch, or there is an available parking stretch ahead while the car is not reaching to the end of the stretch. This testing technique is sufficient to cover all the requirements of this method, identify the system's behaviours then decide the follow-up actions.

Conditions / Actions	TC_P_01	TC_P_02	TC_P_03
c1: Already at valid parking stretch? (Car can park, no need to move forward)	Yes	No	No
c2: Valid stretch found ahead? (Car will move forward and a free parking spot is guaranteed before reaching the end of the stretch)	-	Yes	No
c1: End of street reached? (Car will move forward but no free parking spot is found at the end of the stretch)	-	No	Yes
a1: Move forward	No	Yes	Yes
a2: Parking Status	PARKED	PARKED	UNPARKED

4.5. UnPark

4.5.1. Requirements

Requirements R9.1 and R9.2 (listed in [Section 2](#)) are satisfied by this method.

4.5.2. Inputs

- Queries the car's internal parked status using the WhereIs () method

4.5.3. Outputs

- Updates the car's internal parked status

4.5.4. Assumptions

- The car's length is less than 5m; this space should be enough for the car to unpark.

4.5.5. Pre-Conditions

The following invariants should hold before `UnPark()` is executed:

- The car's position is greater than or equal to 1 but less than or equal to 500

4.5.6. Post-Conditions

The following invariants should hold after `UnPark()` is executed:

- The car should be in an unparked state
- The car's position should remain unchanged i.e. between 100 and 500

4.5.7. Test design technique

A decision table was chosen to design the test cases for `UnPark()`:

Conditions / Actions	TC-UP-1	TC-UP-2	TC-UP-3
c1: $1 \leq \text{position} \leq 500$ c2: <code>carParkingState = PARKED</code>	True True	True False	False -
a1: wrong input/state a2: do nothing a3: <code>carParkingState = UNPARKED</code> , car position remains unchanged	- - X	- X -	X - -

4.6. WhereIs

4.6.1. Requirements

Requirements R10.1 and R10.2 (listed in [Section 2](#)) must be covered through this method.

4.6.2. Inputs

- Access the car's position that is an attribute of the `AutonomousParking` class
- Access the car's status which is an attribute of the `AutonomousParking` class.

4.6.3. Outputs

- Return the updated car's position and parking status through a data type implemented in the project.

4.6.4. Pre-Conditions

- Position must be within the defined range $[0 \leq \text{position} \leq 500]$
- `ParkingStatus` could be either `PARKED` or `UNPARKED`

4.6.5. Post-Conditions

- Updated car's position and parking status.

4.6.6. Test design technique

- Decision tables: The action/output state that will report this method depends on the combination of the variables car's position and parking state. Then, this scenario is best modeled by using a decision table.

Conditions/Actions	TC-WI-01	TC-WI-02	TC-WI-03
<i>c1: Position valid? [0 position 500]</i>	<i>F</i>	<i>T</i>	<i>T</i>
<i>c2: parking status = parked?</i>	<i>-</i>	<i>T</i>	<i>F</i>
<i>a1: wrong input/state</i>	<i>X</i>	<i>-</i>	<i>-</i>
<i>a2: position</i>	<i>-</i>	<i>X</i>	<i>X</i>
<i>a4: status: parked</i>	<i>-</i>	<i>X</i>	<i>-</i>
<i>a5: status: unparked</i>	<i>-</i>	<i>-</i>	<i>X</i>

- Boundary value conditions: We used this technique since one of the variables – car position - that directly affect the output of this method has a range of valid values

Variable	TC-WI-01	TC-WI-02	TC-WI-01
<i>Car position</i>	<i>< 0</i>	<i>100</i>	<i>> 500</i>
<i>Method State</i>	<i>Invalid</i>	<i>Valid</i>	<i>Invalid</i>

5. Test cases implementation & outcomes

5.1. IsEmpty

The test cases defined in the test specification were implemented using JUnit 4. Each test case maps the conditions identified during the test design phase to specific sensor inputs and expected outputs. Mock sensor objects are used to provide controlled sensor data, allowing different valid, invalid, and noisy sensor conditions to be tested independently of real sensor hardware.

5.1.1. TC_IE_01: Both Sensors Valid

Input/Output:

Item	Value
Sensor 1 (5 times query)	{101, 100, 102, 104, 105}
Sensor 2 (5 times query)	{10, 20, 30, 40, 50}
Sensor 1 average	102
Sensor 2 average	30
Expected result	30
Condition	Both sensors valid

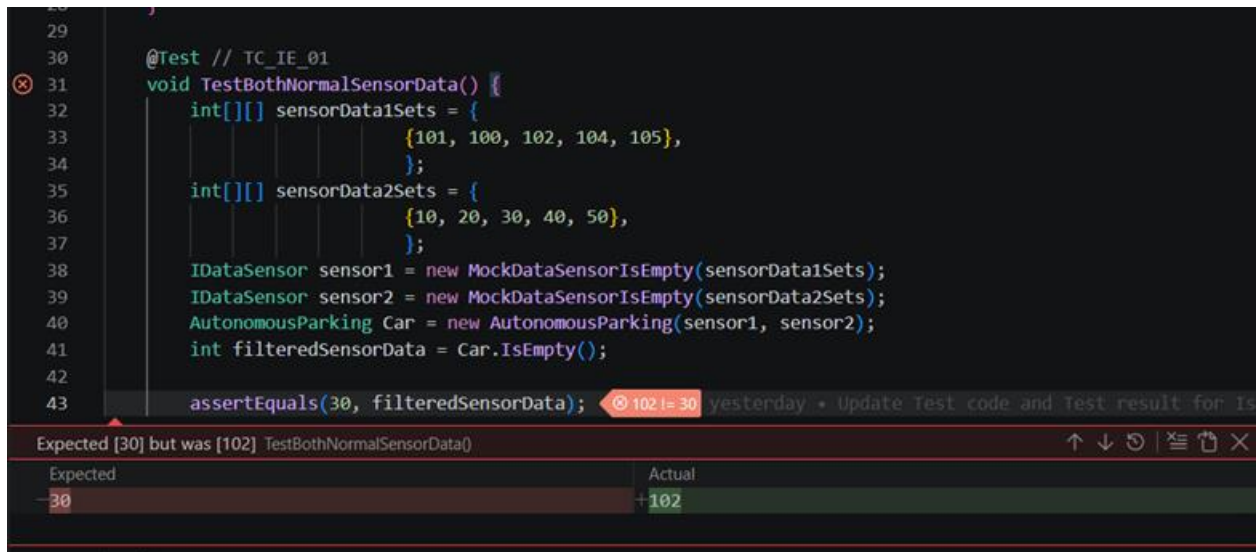
```
@Test // TC_IE_01
void TestBothNormalSensorData() {
    int[][] sensorData1Sets = {
        {101, 100, 102, 104, 105},
    };
    int[][] sensorData2Sets = {
        {10, 20, 30, 40, 50},
    };
    IDataSensor sensor1 = new MockDataSensorIsEmpty(sensorData1Sets);
    IDataSensor sensor2 = new MockDataSensorIsEmpty(sensorData2Sets);
    AutonomousParking Car = new AutonomousParking(sensor1, sensor2);
    int filteredSensorData = Car.IsEmpty();

    assertEquals(30, filteredSensorData);
}
```

Figure 5.1: Initial test for TC_IE_01

5.1.2. Test-Driving Development

a. Test Failure:



```
29
30 @Test // TC_IE_01
31 void TestBothNormalSensorData() {
32     int[][] sensorData1Sets = {
33         {101, 100, 102, 104, 105},
34     };
35     int[][] sensorData2Sets = {
36         {10, 20, 30, 40, 50},
37     };
38     IDataSensor sensor1 = new MockDataSensorIsEmpty(sensorData1Sets);
39     IDataSensor sensor2 = new MockDataSensorIsEmpty(sensorData2Sets);
40     AutonomousParking Car = new AutonomousParking(sensor1, sensor2);
41     int filteredSensorData = Car.IsEmpty();
42
43     assertEquals(30, filteredSensorData);
44 }
```

Expected [30] but was [102] TestBothNormalSensorData()

Expected	Actual
30	102

Figure 5.2: Initial test failure for TC_IE_01

- It resulted in 102 instead of 30

b. Identify the defect



```
/* Determine the final filtered data based on valid sensor readings */
if (filteredDataSensor1 != -1 && filteredDataSensor2 != -1) {
    filteredData = Math.max(filteredDataSensor1, filteredDataSensor2);
}
else if (filteredDataSensor1 != -1) {
    filteredData = filteredDataSensor1;
}
else if (filteredDataSensor2 != -1) {
    filteredData = filteredDataSensor2;
}
```

This should be min()

Figure 5.3: Identify the defect

- Max() was used instead of Min() in the source code.

c. Correct code

```
169
170     if (isSensor2Valid) {
171         filteredDataSensor2 = sensor2.calculateData();
172     }
173
174     /* Determine the final filtered data based on valid sensor readings */
175     if (filteredDataSensor1 != -1 && filteredDataSensor2 != -1) {
176         filteredData = Math.min(filteredDataSensor1, filteredDataSensor2);
177     }
178     else if (filteredDataSensor1 != -1) {
179         filteredData = filteredDataSensor1;
180     }
181     else if (filteredDataSensor2 != -1) {
182         filteredData = filteredDataSensor2;
183     }
```

Change to min()

Figure 5.4: Fix the defect

d. Test again:

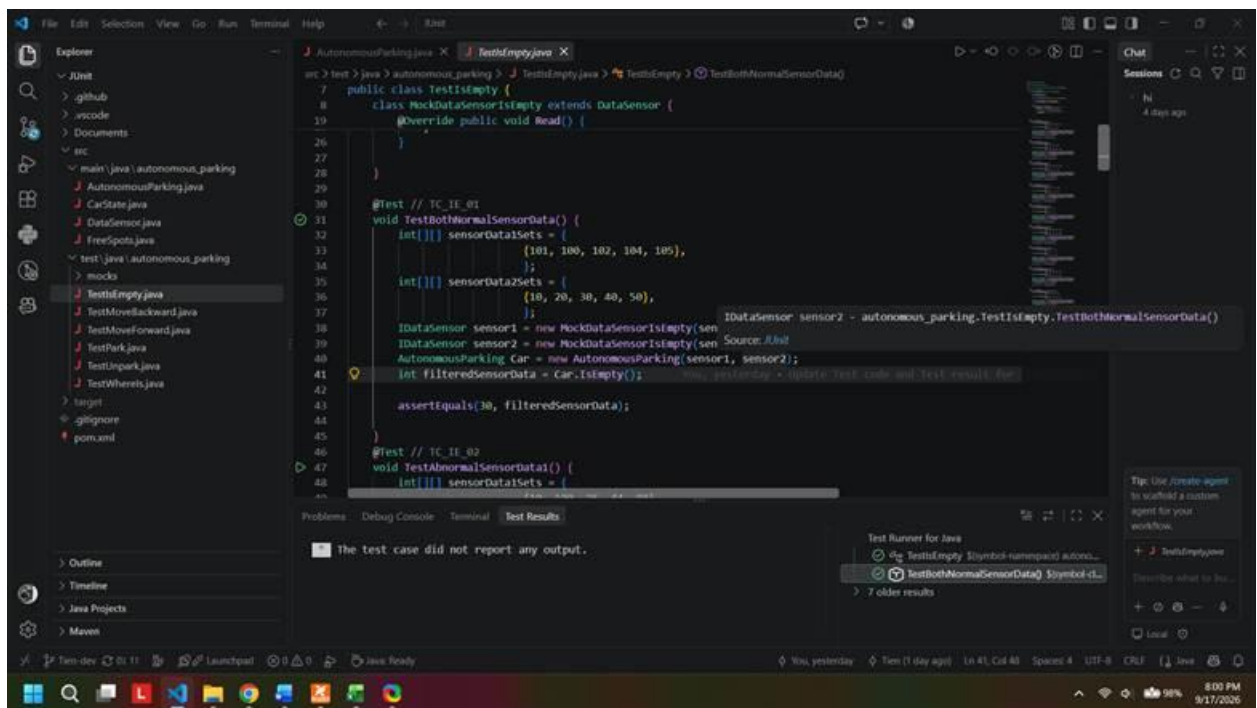


Figure 5.5: Test case is passed due to fixing the code

5.2. WhereIs

The different test cases are derived directly from the Boundary Value condition and the Decision Table. The different test cases are tagged to have a direct relation between the test case and the implemented code.

TAG to keep track
the testing cases

```

@Test //TC-WI-01
void WhereIsRejectsInvalidPosition() {
    AutonomousParking car = new AutonomousParking(
        new MockDataSensor(new int[] {101, 100, 102, 104, 105}),
        new MockDataSensor(new int[] {103, 100, 101, 99, 98}) );
    car.currCarPosition = -1;
    IllegalStateException error = assertThrows( IllegalStateException.class, () -> car.WhereIs() );
    assertEquals( expected: "Invalid car position", error.getMessage());

    car.currCarPosition = 501;
    error = assertThrows( IllegalStateException.class, () -> car.WhereIs() );
    assertEquals( expected: "Invalid car position", error.getMessage());
    System.out.println(error.getMessage());
}

```

Figure 5.6: Test code implementation and TAG to keep track in the class implementation.

5.2.1. Test suit explained

- **TC-WI-01 (Test technique: Boundary conditions + Decision Table):** Simulates the scenario in which the car position is outside of the required range.
- **TC-WI-02 (Test technique: Decision Table):** This test cases verifies the current parking state of the car – provided that the car position is within the limits. If the car is parked, the method returns PARKED as the current state along with the car position. The position must remain unchanged.
- **TC-WI-03 (Test technique: Decision Table):** This test case verifies the opposite situation in which the current parking state is UNPARKED. The method returns UNPARKED as the parking state along with the car position.

5.2.2. Test-Driven Development

```

public CarState WhereIs() {
    return new CarState(currCarPosition, currParkingStatus);
}

```

Figure 5.7: First implementation of WhereIs method.

```
✖ 1 test failed, 2 passed 3 tests total, 1 sec 272 ms
Java\jdk-21.0.12.1\bin\java.exe" ...

AssertionFailedError: Expected java.lang.IllegalStateException to be thrown, but nothing was thrown.
>
> parking.TestWhereIs.WhereIsRejectsInvalidPosition(TestWhereIs.java:15) <1 internal line>
ava.util.ArrayList.forEach(ArrayList.java:1596)
ava.util.ArrayList.forEach(ArrayList.java:1596)
```

Figure 5.7: When test is executed and the car's position is outside the defined range, the test fails.

Included to take care of the range of car's position

```
public CarState WhereIs() {
    if (currCarPosition < 0 || currCarPosition > 500){
        throw new IllegalStateException("Invalid car position");
    }
    return new CarState(currCarPosition, currParkingStatus);
}
```

Figure 5.8: Code included to take care of values outside of range.

```
✓ 3 tests passed 3 tests total, 1 sec 321 ms
"C:\Program Files\Java\jdk-21.0.12.1\bin\java.exe" ...
```

Figure 5.9: Test result after the code improvement.